2 nd Semester

Module - Data Communications and Networks

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1. **How does the physical structure of guided transmission media affect their performance and bandwidth?**

- The physical structure of guided transmission media is the medium which is transmitted signals via physically. The types of guided media are twisted -pair cable, coaxial cable and fiber optic. Guided media is also called as bounded media.

- The physical structure directly affect with the bandwidth of network, easy installation, distance of transmission, signal attenuation. Bandwidth capacity refers the frequencies that can be transmitted per unit of time. Easy installation is flexible in various environment. The distance of transmission is the highest distance which reliability of transformation of signals. The signal attenuation is the difference levels of signal transmission which is efficiency of data transmission.

2. **What are the advantages and disadvantages of using guided transmission media in a network?**

Advantages:

a) High security - It has a higher security than the unguided transmission media. The guided media ia suitable for transmit data than wireless media. Because it was difficult to access for unauthorized users.

b) Reliability - It is more energetic than the unguided media due to the guided media use the physical wire to transmit data instead of unguided media which is the medium through the air.

c) Higher bandwidth - It can be transmit larger amount of data with higher bandwidth rather than the unguided media. It means it is fastest & efficient than unguided media.

d) Cost effectiveness - It is a cost effective data transmission solution which has cheaper to maintain.

Disadvantages:

A) Physical restrictions - Physical restrictions constrain guided transmission media. Maximum transmission distances are limited by cable due to signal attenuation, and longer distances may necessitate signal amplification or repeaters.

b) Limited Mobility - Because guided transmission media depend on physical links, their mobility is limited. Cables are required for connecting devices to the network infrastructure, which can be less flexible and mobile than wireless networks.

C) Scalability Issues - It may be more difficult to change or expand a network that makes use of guided transmission media. In contrast to wireless networks, which can be scaled more quickly, adding new connections or expanding the network frequently requires installing extra cables and can involve complicated design and implementation.

d) Installation and upkeep - When compared to wireless options, guided transmission medium installation can be more difficult and time-consuming. Additional effort and knowledge may be needed for cable running and making sure connections are correct.

3. **Provide examples of real-world applications where guided transmission media moly used**

I. Local region Networks (LANs): LANs connect PCs, servers, switches, and other network devices within a constrained geographic region using Ethernet cables, such as twisted-pair copper cables. These cables offer dependable and fast data transfer, facilitating communication inside of homes, workplaces, and educational institutions.

II. . For long-distance communication connectivity for wide area networks (WANs), fiber optic cables are frequently used. They make it possible for data to be transmitted across great distances and at high bandwidth between scattered locations. A wide area network (WAN) links distant offices, campuses, and even different cities to enable effective data transfer across long distances.

III. Coaxial cables are frequently used in cable television networks to deliver TV signals and give subscribers access to broadband internet. Coaxial cables are ideal for providing television and internet services because they provide high capacity and can transfer signals over relatively long distances while retaining signal quality.

IV. . Telephone networks have historically employed copper wires, most notably twisted-pair cables, for voice communication. These cables offer a dependable way to link phone exchanges, service individual phone lines, and send analog voice signals across great distances.

V. Fiber optic cables are widely utilized in data centers to connect servers, storage systems, and networking devices. They support the huge data flow within these facilities by enabling high-speed data transfer and low-latency communication.

VI. Industrial Control Systems: In industrial settings, control systems, sensors, and devices are connected by guided transmission media, such as fiber optics and industrial-grade Ethernet cables. For automation, monitoring, and process control systems in sectors including manufacturing, energy, and transportation, these media guarantee dependable and secure connection.

VII. . Video surveillance systems rely on guided transmission medium to link cameras to storage or centralized monitoring stations, which is a critical component. Depending on the specific requirements of the monitoring system, coaxial cables or network cables are frequently employed.

VIII. . Home networking: Applications for home networking in domestic settings utilize guided transmission media. To link devices like laptops, smart TVs, game consoles, and routers within a home network and enable data exchange

and internet connectivity, coaxial cables, power line communication adapters, or Ethernet cables are often used.

4. What are the seven layers of the OSI model and their respective functions?

1) Physical Layer-voltage levels, data rates, connectors, and physical topology.
2) Data Link Layer-framing, error detection and correction, flow control, and media access control (MAC).
3) Network Layer-logical addressing, routing, fragmentation, and congestion control.
4) Transport Layer-segmentation, reliable data transfer, flow control, and error detection and recovery.
5) Session Layer-session establishment, synchronization, check pointing, and session termination.
6) Presentation Layer- data translation, encryption/decryption, data compression, and syntax conversion.
7) Application Layer-providing network services to applications, user authentication, and application-specific protocols.

5. **How does the OSI model facilitate communication between different network device and systems?**

- Quick recap of 7 layers of OSI model & their respective functions as follows:

1. Physical Layer: This layer manages physical stuffs on the Computer together. Like managing electronic signals, light signals, radio signals etc. This is the hardware layer of the OSI model.

2. Data Link Layer: This layer simply responsible for encoding and decoding signals into bits.

3. Network Layer: This layers main duty is moving packets from source to destination and create the inter-networking.

4. Transport Layer: This is an important layer because this layer decide how much information is sent at a time.

5. Session Layer: This layer maintains proper communication between two computers. As an ex: when we visit a website dealing with our PC and Server PC.

6. Presentation Layer: This layer operates data with the OS like translation, encryption, and compression.

7. Application Layer: This is the topmost layer where end user is interacting with.

- So, this OSI model is like set of rules and like a road map with different layers. Each of these layers has a special job as I mentioned before. By using this model if devices made by different companies they can easily work together, because every manufacture using the same model.
- This model is also flexible. Why I'm saying that is if we want to change a layer, we can change that without affecting other layers.
- By using this model devices can communicate reliably. They can send and receive data properly. It's like having a clearly organized system that everyone agrees to follow.
- So, OSI model is set of rules that simplify the way of communication between devices and systems. By using that we can organize data communication and that makes compatibility among different devices. It ensures reliable data exchange by following the rules at each layer.
- So that’s how the OSI model facilitates communication between different network devices and systems.

6. **What are the advantages of using the OSI model as a network architecture and design framework?**

- advantages of using the OSI model as a network architecture •provides a common language of reference point between network professionals
    - David's networking tasks into logical layers for easier comprehension
    - Allows specialization of features at different levels
    - Aids in troubleshooting
- •promotes standards of interoperability in networks and devices
- Provides modularity in networking features
- advantages of using the OSI model as a design framework
- Makes easier the Development Process.
- The frameworks provide the tools and packages to help people start developing their websites.

•Eases Debugging and Application Maintenance.

•Reduces Code Length.

•Improves Database Proficiency.

•Reinforces security.

7. **Can you describe the Process of data encapsulation and DE capsulation at each layer of both the OSI model and the TCP/IP model?**

❖ OSI encapsulation

-The Open Systems Interconnection (OSI) model, developed by ISO, is a paradigm for protocols that divides system levels according to function. The seven OSI layers each offer a distinct set of services.

A source computer transmits a packet from Layer 7, the application layer, to Layer 1, the physical layer, as part of the encapsulation process. A packet doesn't start encapsulating data until it reaches Layer 4, the transport layer. The packet is then given the necessary information by the following layers, allowing it to move through the system.

As the packet is transmitted to the recipient computer, DE capsulation occurs in the OSI model's reverse order, from Layer 1 to Layer 7.

First layer: Physical

This is the layer that controls the electrical impulses sent by an active network device. The bit sequence that denotes the start of the packet and the trailer is added by the physical layer.

Data link at layer

The link layer controls communication between transmitting and receiving systems via the local network. The local destination address is added to the header during encapsulation.

Network, Layer

The packet is forwarded across the network via the network layer. The packet's internet address header is encapsulated.

Transportation

Transmission Control Protocol (TCP), among other transport protocols, provides assurance that all packets will arrive at their intended location. User Datagram Protocol (UDP) and other transport protocols do not ensure delivery.

Session

The information transfer between computers is controlled by the session layer. It offers details on how information is permitted to move freely throughout the system.

Presentation, layer six

The information is changed by the presentation layer so that it can be used by an application. It offers data formats information.

Application

The application layer, which is the top layer, communicates with the end device.

- OSI DEcapsulation

a. Physical Layer: At this layer, there is no decapsulation process because it deals with the physical transmission of data over the network medium, such as cables or wireless signals.

b. Data Link Layer: The decapsulation process at this layer involves removing the data link layer header and trailer from the received frame. It extracts the data portion and passes it to the network layer.

c. Network Layer: Here, the decapsulation process includes removing the network layer header from the packet. It retrieves the encapsulated IP packet and forwards it to the transport layer.

d. Transport Layer: At this layer, the decapsulation involves removing the transport layer header from the segment. It retrieves the encapsulated segment and passes it to the session layer.

e. Session Layer: The decapsulation process at this layer primarily focuses on establishing, managing, and terminating sessions between applications. It does not involve decapsulation in the traditional sense.

f. Presentation Layer: Similarly, the presentation layer mainly deals with data formatting, encryption, and compression. It does not perform decapsulation.

g. Application Layer: The decapsulation process at this layer involves removing the application layer header and extracting the payload data, making it available to the application.

TCP encapsulation

The Internet Protocol suite, often known as TCP/IP and created with funding from the US Department of Defense, contains the most extensively used network protocols. The TCP/IP model was created to function on various kinds of computers.

The OSI model and TCP/IP models are different even if they both adhere to the same design principles. The OSI model's Layers 5 through 7 are combined into a single application layer in the TCP/IP model, which is the primary distinction. A layer sometimes referred to as the network interface or network access layer, it also sometimes includes the physical and data link levels.

Similar to the OSI paradigm, the TCP/IP encapsulation and DE capsulation procedures progress via the layers.

Data link and physical interface layers of a network

Each type of network medium's unique bit sequence is added to the packet by the lowest layer, the physical layer, which also encapsulates it. Ethernet, fiber, and Wi-Fi networks, for example, operate differently from one another. As a result, each has a unique set of bytes that indicate the start of a packet. By type of medium, the trailer format also differs.

Every packet travels across the local network for the first hop. The media address control (MAC) identification, which the system manufacturer installs in each network interface, is added by the data link layer. Every network interface that the manufacturer has ever made is uniquely specified in the following bytes, which come after the first bytes that identify the company. As a result, no two interfaces on earth have the same MAC address.

Every system connected to a local network examines a packet's destination MAC address when it is sent. However, the remainder of the packet is only read by the machine that has that particular allocated MAC address.

Ethernet layer

The packet is routed through the deeper network by the network layer. A router is frequently reached by a packet during its initial hop across the local network, which is handled by the data link layer.

To address the target system across the wider network, the packet needs an internet address. The destination internet address, which is given to a system when it is installed in the network, is added to the packet header by the network layer.

Conveyance layer

There are numerous transport protocols available, including Stream Control Transmission Protocol (SCTP), TCP, and UDP:

TCP ensures packet delivery by requiring that every packet be acknowledged upon receipt. It updates the header with the total number of bytes sent in the packet before waiting for the recipient to reply with the number it is acknowledging. In order to avoid the sender from overpowering it by delivering data too quickly, the receiver likewise adds the amount of bytes that it is now prepared to accept.

Despite the fact that UDP does not ensure delivery, it is a useful protocol for latency-sensitive applications like speech or video because dropping a packet is preferable to taking the time to resend lost ones.

SCTP has the ability to transfer data and control information in independent, concurrent streams.

Applying layer

The session and presentation levels are combined into the application layer in the TCP/IP architecture. A connection's session can decide whether it transmits in both directions at once, one direction at a time, or only one direction.

The presentation layer, which runs over sessions, makes sure the application receives data in a format that is useful. Along with compression and decompression, it can also provide encryption and decryption.

The human interface to the network, including features like email, file sharing, and the web, is defined by the application layer.

Conclusion

Although they may be improved, current protocols won't be fundamentally altered to maintain compatibility. Network professionals should anticipate the emergence of new protocols in the future that will probably maintain the layered structure. Additionally, even while specifics alter, the idea of encapsulation does not.

- TCP Decapsulation

TCP/IP Model: a. Network Interface Layer: The physical and data connection layers of the OSI model are similar to this layer in the TCP/IP model. The data link layer of the OSI model describes the decapsulation method in the same manner.

b. Internet Layer: Comparable to the network layer in the OSI model, decapsulation entails stripping the IP header from the encapsulated packet in order to extract it and send it on to the transport layer.

c. Transport Layer: The decapsulation procedure is the same as that outlined for the transportation layer in the OSI model, where the transport layer headers is eliminated to obtain the encapsulated segment and pass it to the higher levels.

d. Application Layer: Similar to the OSI model's application layer, which removes the application surface header, the decapsulation method removes the application layer.

The program is given access to the payload data at position d.

It's vital to remember that the TCP/IP paradigm, which condenses numerous layers into fewer ones, is frequently regarded as a condensed version of the OSI model. Therefore, although having distinct layer names and organizational frameworks, the decapsulation process in both models adheres to the same basic principles.

Physical tier: Because it deals with the actual physical transfer of data through the network media, such as cable or radio waves, there is no decapsulation procedure at this tier of the OSI model.

b. Data Link Layer: The incoming frame's data link layer header and trailer are taken out throughout this layer's decapsulation operation. The data part is extracted, and the network layer receives it.

c. Network Layer: In this case, the packet's network layer header is removed as part of the decapsulation procedure. It gets the IP packet that has been encapsulated and sends it on to the transport layer.

d. Transport tier: Decapsulation at this tier entails deleting the segment's transport layer header. It gets the part that has been encapsulated and sends it to the session layer.

8. **What is the TCP/IP model, and how does it differ from the OS model?**

A group of communication protocols called TCP/IP, or Transmission Control Protocol/Internet Protocol, are used to connect network devices on the internet. An intranet or extranet is a type of private computer network that uses TCP/IP as its communication protocol.

TCP/IP is a general term for the complete IP suite, a collection of protocols. Although there are other protocols in the suite, TCP and IP are the two most common. The TCP/IP protocol suite serves as an abstraction layer between the routing and switching fabric and internet applications.

TCP/IP only has four layers, but OSI has seven.

Systems that are open to interconnection and communication with other systems employ the OSI Model, which defines network communication. The TCP/IP protocol, on the other hand, enables you to choose how a particular computer should be connected to the internet and how data can be sent between them.

TCP/IP header size is 20 bytes, compared to 5 bytes for the OSI header.

TCP/IP stands for Transmission Control Protocol, whereas OSI stands for Open Systems Interconnection.

TCP/IP uses a horizontal method, whereas OSI uses a vertical approach.

Unlike the TCP/IP paradigm, which is both connection-oriented and connectionless, the transport layer of the OSI model is simply connection-oriented.

9. **What are the four layers of the TCP/IP model and their respective functions? 10. How does the TCP/IP model relate to the protocols and technologies used in the internet?**

Layer of Network Access

This is the TCP/IP model architecture's lowest layer.

It combines the OSI model's Data Link and Physical Layers.

At this layer, actual data transmission happens.

Encapsulating the IP datagram into the frames after they have been transferred by a network occurs in this layer.

Additionally, this is where the IP address to physical address mapping is done.

This layer's primary purpose is to transfer data between two devices linked to a network

Internet Layer 2.

It is the second layer of the TCP/IP model, and in terms of structure, this layer is comparable to the Network Layer of the OSI Model.

The Internet layer's primary duty is to send data packets to their destination networks.

At this level, logical data transmission happens.

This layer employs three separate protocols. These consist of:

IP: One of the most crucial protocols since it identifies a device's IP address, which is then used for network connections. The method through which the data will be transmitted is chosen using this protocol.

ARP: It stands for Address Resolution Protocol. The physical address from the IP address can be determined using ARP.

ICMP: It stands for Internet Control Message Protocol and notification regarding datagram problems can be sent back to the user using this. Any issue with the network is immediately notified to the user by ICMP. It can only inform the user about the errors and cannot rectify the problem

3. Host-To-Host Layer

This layer is parallel to the transport layer of the OSI Model

The error-free delivery of data is the main function of this layer

There are two main protocols present in this layer:

TCP: Another integral part, the Transmission Control Protocol is a reliable communication protocol. It manager the flow of data, i.e. the sequence and segmentation of the data

UDP: It is a connection-free protocol which makes it cost-effective but less reliable.

4. Application Layer

The topic three layers of the OSI Model: Application, Presentation and Sessions, when combined together, they perform similar functions as the Application Layer of the TCP/IP model

Node-to-node communication based on the user-interface occurs here

Multiple protocols are present in this layer, a few common ones have been mentioned below in brief:

HTTP: Hypertext Transfer Protocol is used to manage the communication between the server and web browsers

NTP: Network Time Protocol can set one standard time source in our computer, which enables sync between the server and the user

TELNET: Telecommunication Network is used to have access to files present of the Telnet network and manage them on internet

FTP: File Transfer Protocol, as the name suggests allows easy transferring of files

10. .How does the TCP/IP model relate to the protocol and technologies used in the internet

- The TCP/IP model is like a plan that helps devices communicate over the Internet. It has four layers: Network Interface, Internet, Transport, and Application.

- The Network Interface layer connects devices physically to the Internet.
- The Internet layer helps data find its way across different networks.
- The Transport layer ensures data is delivered correctly and in order.
- The Application layer includes programs like web browsers and email apps.

- Protocols and technologies used in the Internet follow this model. They work together to ensure data is sent and received properly.

- By using the TCP/IP model, devices from different manufacturers can communicate. It's like everyone speaking the same language and following the same rules. This makes it easier for data to flow smoothly over the Internet.